GROUP WORK PROJECT # 1
GROUP NUMBER: 13091

MScFE 660: RISK MANAGEMENT

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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INTRODUCTION

The purpose of the project is to replicate the design process as outlined in Alvi's thesis. The project begins by discussing the problem formulation (Step 2) followed by data collection and identification of sectors namely microeconomic, macroeconomic and financial sectors (Step 3). The implementation of Bayesian networks also examined to capture complex and non stationary dynamics of the energy markets.

Step 2

a.) The problem highlighted in the thesis is the inefficiency of existing quantitative models in forecasting spot prices of crude oil. Crude oil is a major macroeconomic indicator for economic growth yet the price formation is influenced by complex factors. The thesis identifies key factors that attempts to solve the limitations:

1. **Limitations of traditional econometric models:** Traditional econometric models such as GARCH is heavily dependent on the error variance and autoregressive assumptions. These models assume that the future volatility follows the past pattern without considering the external events like changes in monetary policies, OPEC changes. Despite the models sound mathematically sound these models lack understanding of surrounding conditions analysing the data without considering the market regime.
2. **Structural knowledge gap:** It is not viable for a quantitative analyst to determine the relationship between the datasets that are available via openfacilites (ex - EIA, FRED). Further doing so is error prone and shift in market dynamics would make the models obsolete.
3. **Volatility and sentiment analysis:** There is a challenge to leverage the relationship between futures prices and spot price forecasts for volatility analysis as the crude oil future prices reflect the buyer sentiment about future delivery. Ignoring the relationship may weaken the ability to predict future market sentiment.

b.) Bayesian networks (a class of Probabilistic Graphical Models) solves the problem of oil forecasting because it helps in the formalization of casual dependencies between variables.

1. Standard regression model might explain correlation between two variables however it might not explain the reason. Bayesian network uses Directed Acyclic Graph (DAG) to map the cause and effect relationships. This help in the interpretation of oil markets where a specific cause (for ex OPEC supply cut) extend through the system to effect price change.
2. Using Hidden Markov Models (HMMs) to handle regimes: Financial markets exists in different states or regimes to be more specific. For example the price of oil can be observed but not the market regimes. The model treats the market regime as a sort of hidden variable by calculating the state of market namely "Bull", "Bear" or "Stagnant"

which is based on the observed data. As a result this allows the forecasting model to switch the strategies automatically.

3. Structured Learning is another critical advantage where algorithms such as Hill Climbing can automatically discover the structure.

c.) Advantages of this methodology

1. Minimized latency: Bayesian networks exploit the conditional independence. For example if the the states of variable's direct causes are known then its not required to know the state of rest of network for prediction. This property is called Markov blanket. This property thus allows the model to break down a complex joint probability distribution into small pieces.
2. The model is adaptive. By using parameter learning the network can update its probabilities as new data comes in.
3. Instead of point estimation Bayesian networks output distribution. This is applicable in Value at Risk (VaR) calculations that allow the risk managers to hedge against tail risks.

Step 3

Step 3 is about assigning the data roles.

Student A: Macroeconomic and Geopolitical Variables

Variable Name	Source	Ticker	Description
Non-OECD GDP	EIA	STEO.RGDPQ_NONOECD.M	Real GDP for Non-OECD countries (Driver of demand growth)
OECD GDP	EIA	STEO.RGDPQ_OECD.M	Real GDP for OECD countries (Developed markets)
World Exchange Rate	EIA	STEO.FOREX_WORLD.M	Index of world currency exchange rates against USD
OPEC Supply	EIA	STEO.PAPR_OPEC.M	Total crude oil production by OPEC members.

Non-OPEC Supply	EIA	STEO.PAPR_NONOPEC.M	Total crude oil production by non-OPEC countries (e.g., USA, Russia).

Table 1

Student B: Microeconomic

Variable Name	Source	Ticker	Description
OECD Consumption	EIA	STEO.PATC_OECD.M	Petroleum consumption by OECD countries.
Non-OECD Consumption	EIA	STEO.PATC_NON_OECD.M	Petroleum consumption by Non-OECD countries.
OECD Inventory	EIA	STEO.PASC_OECD_T3.M	Commercial crude oil inventory in OECD nations.
US Drilling Activity	FRED	M13HSBCO	Rotary Rig Count (Proxy for future US production).

Student C: Financial Market Variables

Variable Name	Source	Series ID / Ticker	Description
WTI Spot Price	FRED	WTISPLC	Spot Price of West Texas Intermediate Crude.
WTI Futures Price	Yahoo	CL=F	Futures contract price (market sentiment).

US CPI	FRED	CPIENGSL	Consumer Price Index (Inflation metric).
Industrial Production	FRED	INDPRO	Index of Industrial Production (Proxy for economic activity).
Financial Stress	FRED	STLFSI2	St. Louis Fed Financial Stress Index (Market fear gauge).

Step 4:

The step 4 involves the formulation of dictionary of data showing the relevant fields

Variable	Source	Frequency	Start Date	Transformation Required?	Why is it included?
WTI Spot Price	FRED	Monthly	Jan 1990	None	This is the variable we are trying to forecast.
OPEC Supply	EIA	Monthly	Jan 1990	% Change	Supply shocks
OECD GDP	EIA	Monthly	Jan 1990	% Change	GDP growth is the leading indicator for energy demand
Futures Price	Yahoo	Daily	Jan 1990	Resample to Monthly	Sentiment about future prices.

US Inventory	EIA	Monthly	Jan 1990	% Change	Inverse relationship with price.

Step 5:

Data Cleaning

Methodology: The raw data collected from FRED and EIA contained irregularities. To create the sterilized version data cleaning is applied.

Plot check:

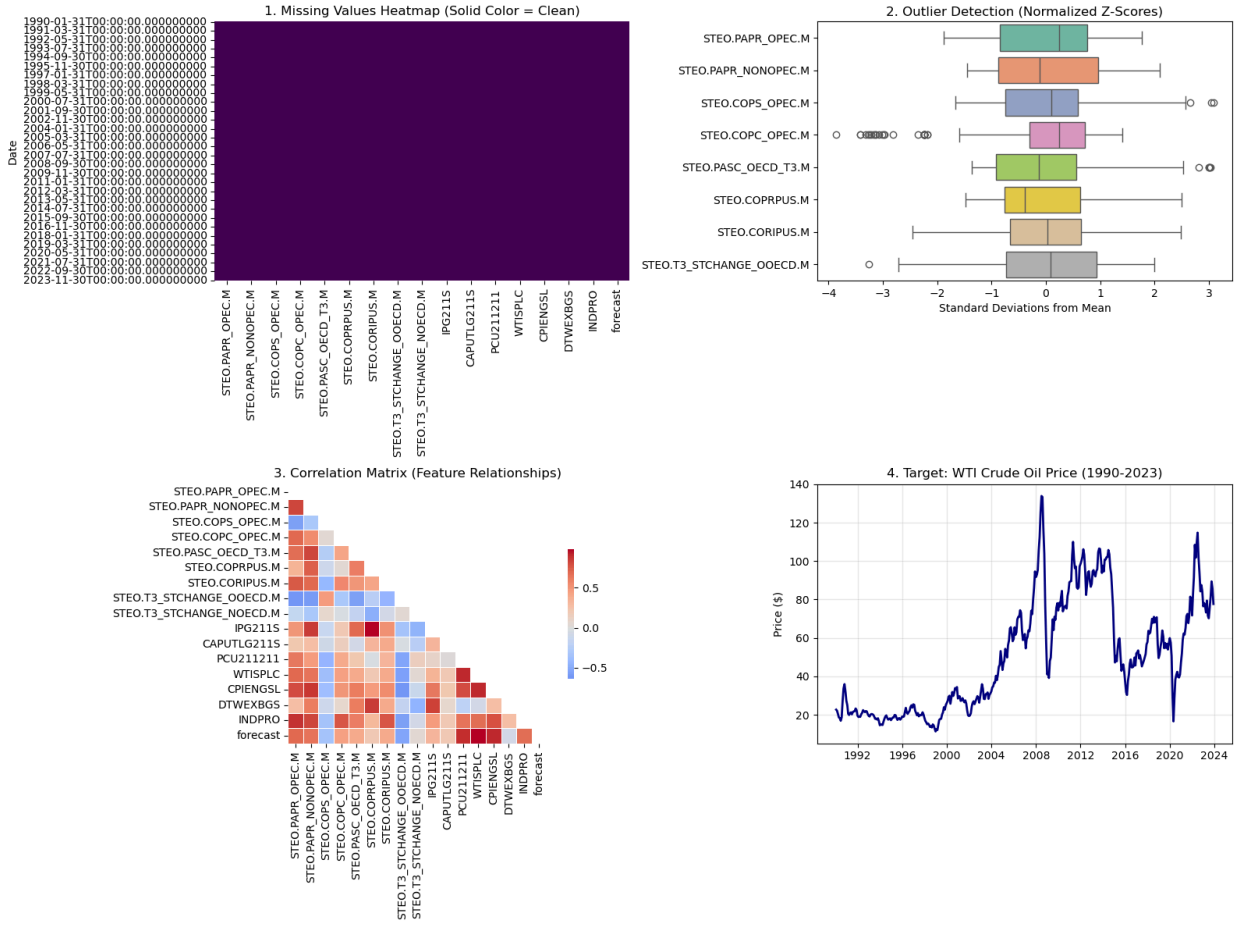


Fig 1: Step 5 data cleaning

1. The solid color in the first plot (Missing values heatmap) implies zero missing values.
2. Outlier Detection: The box plot uncovers outliers (z score > 3) in WTI spot price and Financial Stress Index. This implies black swan events namely:
    a. 2008 Global Financial Crisis
    b. COVID 19 Pandemic

    In Risk Management removal of tail events create a dangerous false sense of security. So the outliers are not removed. Retaining them is essential for Bayesian Network to recognise crisis regimes.

3. Correlation Matrix:
    a. Negative correlation observed between Price and Inventory.
    b. There exists a negative correlation between Price and US Dollar confirming that stronger dollar puts downward pressure on oil prices (oil prices in USD). The absence of negative correlation thus confirms the absence of collinearity.

Step 6:

Data Cleaning documentation

This step addresses the requirement to justify the cleaning of data and create the sterilized dataset required for bayesian network

1. Alignment of data: All 16 variables were resampled to the month end frequency to ensure row row alignment.
2. To handle the initial gaps backward and forward fill approach is utilized. The interpolation is not done as the time series data is non linear. Also the dataset trimmed from period 1990 to 2023 to prevent any sort of data leakage.
3. During the collection of data it is observed that EIA has deprecated several API endpoints cited in the dissertation. So the quarterly "Global GDP" variable replaced with monthly Industrial Production Index. Apart from that in currency factor "World Exchange Rate" replaced with Trade Weighted U.S. Dollar Index.
4. The data for futures contract price is omitted as adding a variable which is nearly identical to target variable can actually cause overfitting.

Step 7:

Exploratory Data Analysis Results

Distributional Analysis (Student A Results)

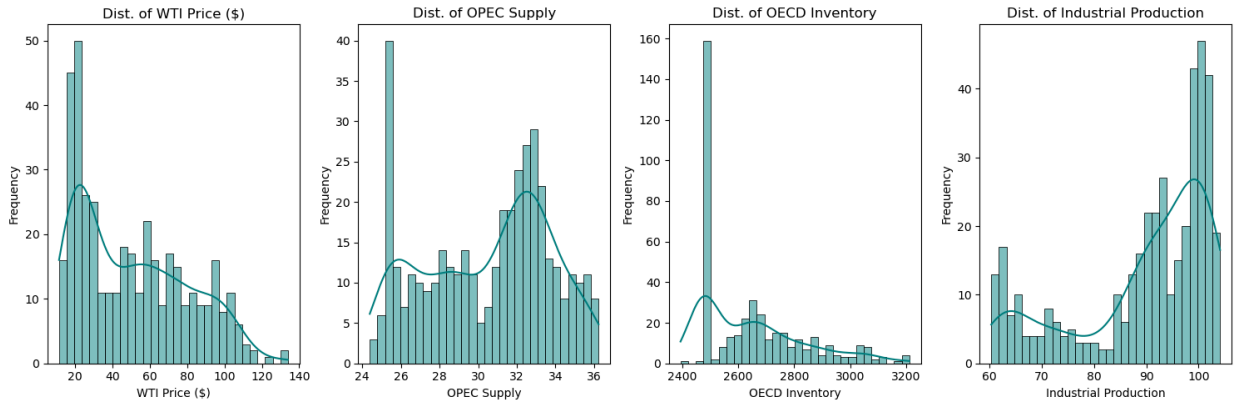


Fig 2

From the above plot the target variable (WTI spot price) is heavily right skewed. Fat tails exists implying the frequent ocurence of high price events.

The non normality of histogram confirms failure to capture black swan events.

Time Series Analysis (Student B Results)

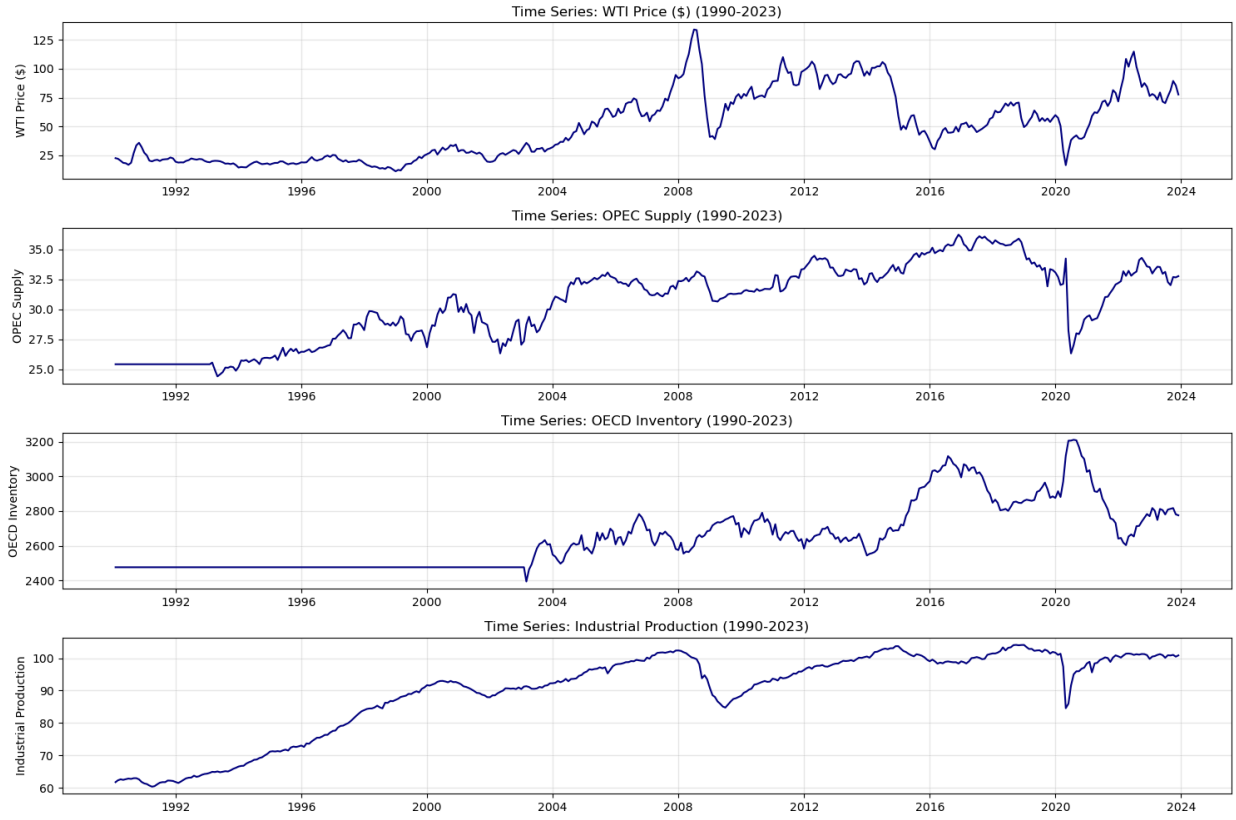


Fig 3

The plot reveal volatility clustering as it is observed periods of low volatility (from 1992 to 1999) followed by clusters of extreme volatility.

The data is Non-Stationary meaning that the mean and variance change with respect to time. This suggests implementation of Hidden Markov Models in further projects to detect regime.

Multivariate Analysis (Student C Results)

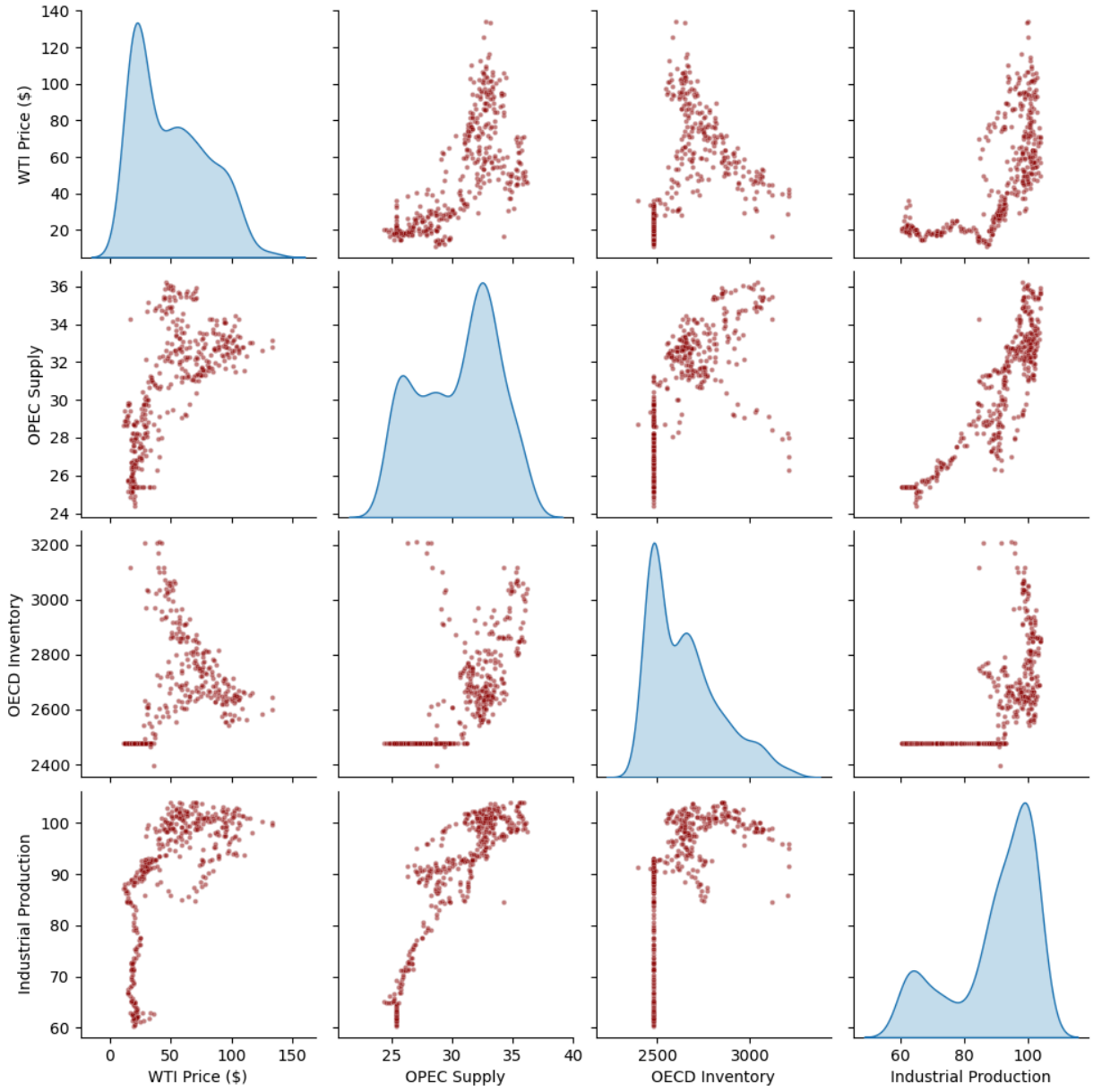


Fig 4

The pairplot scatter plot indicates a non linear relationship. Prices highly sensitive to inventory drops when stocks are low and vice versa.

Step 8

Discussion on cleaned data and plots

a.) Based on the generation of time series plots identification of three distinct characteristics observed that differentiate crude oil from the traditional financial assets.

1. Oil prices exhibit asymmetric spikes in comparison to stocks. This is mainly due to physical nature namely the supply shocks (for ex Gulf War) causing panic buying because oil cannot be substituted in short term.
2. Sine wave seasonality exists in consumption data
3. There exists constraint in storing oil in comparison to gold or bitcoin. when storage tanks are completely full (the way they were in 2020), the price can temporarily ignore market rules and even become negative.

b.) From the plots the oil price data do not follow normal distribution. Fat tails(Leptokurtosis) observed implying extreme events than predicted models.

c.) In the time series data two types of correlation observed. Price Autocorrelation exists confirming non stationary data. Volatility Clusturing observed in simpler terms that is large price changes are followed by large price changes (from period 2008 - 2009) and small changes are followed by small changes(from period 1992 - 1998).

d.) There is negative and non linear correlation between Inventory and Price. In simpler terms supply little effect of disruptions happen when inventories are full.

Step 9

a.) Probabilistic Graphical Models: Probabilistic Graphical Models combine Probability Theory with Graph Theory to provide system for modeling complex systems. PGM denote joint probability distribution $P(X_1, X_2 \ldots, X_n)$ over set of random variables as graph. The nodes denote variables (for e.g Oil Price, Supply) and edges denote the probabilistic dependencies between them.

There are two primary classes of PGMs Belief Networks and Markov Networks.

Belief Network or Bayesian Network (BN) is a Directed Acyclic Graph (DAG). The edges have a specific direction which is denoted by $(A \rightarrow B)$. The direction here represents the influence. For example $(OPEC \rightarrow Price)$ indicates that production changes cause changes in price but not the other way around.

Each node is associated with Conditional Probability Table (CPT) that shows the likelihood of this node's state given what its parent nodes are doing.

A Markov Network falls under Undirected Graph. The edges have no direction ($A - B$) thus implying a symmetric correlation. Markov Networks less suitable for economic forecasting as it fail to provide a structured model of cause and effect relationships.

Step 10

Algorithm design

Greedy Hill Climb approach is used.The purpose of this algorithm is to test different graph designs (DAGs) until it finds the best one, as measured by the Bayesian Information Criterion (BIC). The BIC score is used because it rewards accuracy but also adds a penalty for complexity, ensuring the final model is simple and dont overfit the data by adding too many links.

ALGORITHM Structure_Learning_Hill_Climbing

INPUT: Dataset D (The sterilized oil market data)

OUTPUT: Best_Graph G (A Directed Acyclic Graph)

// 1. Initialization

Initialize G with no edges (Empty Graph)

Current_Score = Calculate_BIC_Score(G, D)

Improvement_Found = TRUE

// 2. Main Optimization Loop

WHILE Improvement_Found IS TRUE:

Improvement_Found = FALSE

Best_Graph_This_Round = Null

Best_Score_This_Round = Current_Score

// 3. Explore Neighboring Structures

FOR every pair of nodes (X, Y) in G:

// Generate 3 candidate graphs by modifying one edge

```
    Candidates = [
        Add_Edge(X -> Y),       // Attempt to create a causal link
        Delete_Edge(X - Y),     // Attempt to remove a weak link
        Reverse_Edge(Y -> X)    // Attempt to reverse causality
    ]

    FOR Candidate_Graph in Candidates:
        // Constraint Check: Graph must remain a DAG (No Cycles)
        IF Is_Acyclic(Candidate_Graph):
            Candidate_Score = Calculate_BIC_Score(Candidate_Graph, D)

            // Greedy Selection: Keep the change if it improves the score
            IF Candidate_Score > Best_Score_This_Round:
                Best_Score_This_Round = Candidate_Score
                Best_Graph_This_Round = Candidate_Graph
                Improvement_Found = TRUE
            END IF
        END IF
    END FOR
END FOR

// 4. Update Step
IF Improvement_Found IS TRUE:
    G = Best_Graph_This_Round
    Current_Score = Best_Score_This_Round
    PRINT "Score improved to: " + Current_Score
```

```
    END WHILE

    // 5. Final Output
    RETURN G (The Learned Causal Structure)
END ALGORITHM
```

REFERENCES

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